Politecnico di Torino

Cybersecurity 01DEHJM

AI-Based Anomaly Detection in Network Security

Alp Kale
S325632
s325632@studenti.polito.it

Ege Kaya
S326635
s326635@studenti.polito.it

Dogan Ege Bulte
S325192
s325192@studenti.polito.it

Contents

1 Introduction **4**
1.1 Problem and Solution Statement 4
1.2 Paper Organisation 4

2 Background **4**
2.1 Anomaly Detection Systems 4
2.2 Types of Network Anomalies 5
2.3 Traditional Detection Tools 5
2.3.1 Snort 5
2.3.2 Suricata 5
2.4 Limitations of Traditional Systems 6

3 AI-Based Anomaly Detection **6**
3.1 Core Principle 6
3.2 Detection Pipeline 6
3.3 Advantages Over Signature-Based Systems 7
3.4 Challenges and Limitations 7

4 AI and Machine Learning Methodologies **7**
4.1 Classical Machine Learning 8
4.1.1 Isolation Forest 8
4.1.2 One-Class Support Vector Machines 8
4.1.3 Random Forest and Ensemble Methods 8
4.2 Deep Learning Approaches 8
4.2.1 Autoencoders 8
4.2.2 Recurrent Neural Networks and LSTM 9
4.2.3 Convolutional Neural Networks 9
4.3 Clustering-Based Approaches 9
4.4 Comparison of Methodologies 10

5 Applications, Discussion, and Mitigations **10**
5.1 Benchmark Datasets and Evaluation 10
5.2 Enterprise Network Security 11
5.3 National Defence and Military Networks 11
5.4 Critical Infrastructure Protection 11
5.5 Advanced Persistent Threat Detection 12
5.6 Mitigation Strategies and Best Practices 12
5.7 Open Challenges 13

6 Conclusion **13**

Abstract

Traditional rule-based intrusion detection systems can only identify attacks they have already seen, leaving networks exposed to zero-day exploits, novel malware, and advanced persistent threats. This paper examines how artificial intelligence and machine learning address this gap through anomaly-based detection: learning a model of normal traffic and flagging deviations, without requiring prior knowledge of specific attacks.

We begin with the structure and documented limitations of traditional Network Anomaly Detection Systems, using Snort and Suricata as concrete examples. We then describe the core principles of AI-based detection and the typical detection pipeline. The paper surveys the main machine learning and deep learning methodologies, including Isolation Forest, One-Class Support Vector Machines, autoencoders, Long Short-Term Memory networks, and clustering algorithms. The final section discusses deployment contexts ranging from enterprise security operations to national defence and critical infrastructure, together with practical mitigation strategies. Our analysis finds that hybrid systems combining signature-based and anomaly-based detection, supported by continuous retraining, offer the most reliable defence against an evolving threat landscape.

Keywords: network anomaly detection, intrusion detection, machine learning, deep learning, autoencoders, LSTM, cybersecurity, critical infrastructure

1 Introduction

Modern networks underpin almost every aspect of daily life, from industrial control systems and financial infrastructure to consumer devices and cloud platforms. As this connectivity has grown, so has the attack surface. Networks now face sustained threats including Distributed Denial-of-Service (DDoS) floods, reconnaissance scanning, covert data exfiltration, credential attacks, and long-running intrusion campaigns by well-resourced adversaries. Defending against this range of threats requires detection systems that are both broad in coverage and fast enough to operate at line rate.

For decades, the dominant approach to network intrusion detection has been signature-based analysis: a detection system is given a database of known attack patterns and raises an alert when incoming traffic matches one of them. This approach works well for known threats and is computationally cheap, but it is fundamentally reactive. It can only detect attacks that have already been observed, catalogued, and turned into rules. Zero-day vulnerabilities, new malware variants, and deliberately crafted evasion techniques pass through undetected because they produce no pattern that the rule database can match.

Anomaly-based detection offers a different perspective. Instead of asking whether traffic matches a known attack, it asks whether traffic deviates significantly from what has been observed as normal. This framing allows previously unseen attacks to be flagged on the basis of their behaviour alone, with no prior knowledge of their specific form. Machine learning provides the tools to formalise what "normal" means, to measure deviation from it, and to do so at the scale and speed that production networks require.

1.1 Problem and Solution Statement

This paper investigates the gap between traditional, rule-based Network Anomaly Detection Systems and the demands of a dynamic threat environment. We examine why signature-based systems fail against novel attacks, describe how AI-based detection systems establish and exploit a behavioural baseline, survey the principal machine learning and deep learning methods applied to this problem, and discuss real-world deployment contexts including national defence, critical infrastructure, and enterprise security operations.

1.2 Paper Organisation

Section 2 covers background material on traditional detection systems, the main categories of network anomaly, and the documented limitations of signature-based tools. Section 3 describes the principles of AI-based anomaly detection and how the detection pipeline operates. Section 4 surveys the key machine learning methodologies. Section 5 discusses applications, national defence scenarios, mitigation strategies, and open challenges. Section 6 concludes the paper.

2 Background

2.1 Anomaly Detection Systems

A Network Anomaly Detection System (NADS) monitors network traffic continuously and raises alerts when it observes activity that departs from expected behaviour. Most deployed systems are passive: they observe traffic and report suspicious events for human review. A

related class, the Network Anomaly Prevention System (NAPS), also takes automated countermeasures such as dropping malicious flows or isolating compromised hosts. The distinction matters in practice. Passive detection supports human-driven incident response, while active prevention enables near-real-time containment at the cost of potentially disrupting legitimate traffic if the system misfires.

2.2 Types of Network Anomalies

Four categories account for the majority of threats that detection systems must address.

Unusual traffic volume. Sudden, sustained spikes or drops in connection rates, bandwidth, or packet counts often signal an attack. A DDoS campaign floods a target with traffic well above its normal operating range. Conversely, a sharp drop in traffic from a normally active host may indicate that it has been taken offline after a compromise.

Data exfiltration. Attackers who have established a foothold must move stolen data out of the network without triggering alarms. DNS tunnelling encodes payload data inside DNS queries and responses, exploiting the fact that DNS traffic is rarely inspected in depth. ICMP tunnelling embeds data in ping packet payloads. Both techniques produce subtle statistical irregularities in traffic patterns that signature-based tools cannot detect.

Unauthorised access attempts. Brute-force attacks generate large volumes of failed authentication requests. SQL injection appears as anomalous character sequences in HTTP traffic. Zero-day exploits produce connection patterns entirely absent from any signature database.

Network reconnaissance. Before launching an attack, adversaries map the target. Port scanning probes a host across many ports in a short time window, creating a distinctive signature in connection logs. Host discovery sweeps generate traffic volumes inconsistent with normal operations, and automated scanners exhibit regular inter-packet timing that distinguishes them from human-driven activity.

2.3 Traditional Detection Tools

The two most widely deployed open-source network intrusion detection tools are Snort and Suricata, which together account for a dominant share of production deployments.

2.3.1 Snort

Snort was originally developed by Sourcefire and is now maintained by Cisco. It operates as a packet-level rule engine, reassembling TCP streams and applying user-defined rules to their content. A rule specifies a protocol, address and port ranges, and one or more content patterns; when all conditions are satisfied, the rule fires. Snort's main strength is the maturity of its rule ecosystem. Community and commercial rule sets provide broad coverage of known attacks. Its limitation follows directly from the rule-based model: coverage is bounded by what has already been observed and encoded into a rule.

2.3.2 Suricata

Suricata is developed by the Open Information Security Foundation (OISF). It follows the same rule-based model but extends it with multi-threaded processing, built-in application-layer protocol detection, and support for TLS certificate inspection. It produces structured JSON log output that captures flow metadata, DNS transactions, and HTTP headers, making it well-suited

for integration with SIEM platforms. Like Snort, Suricata also supports an inline prevention mode.

2.4 Limitations of Traditional Systems

The core limitation of signature-based detection is that its effectiveness is inversely proportional to the novelty of the attack. Rules are written after a threat is discovered and analysed, so there is always a window during which a new attack spreads undetected. This is the zero-day problem. Even known attack families can evade signature matching through simple obfuscation: changing a single byte in a payload, encrypting traffic, or fragmenting a session across packets can invalidate a rule that was accurate against the original variant.

ICMP tunnelling is a well-documented blind spot for both tools. When an attacker encodes data inside standard ping packets using common penetration testing frameworks, neither Snort nor Suricata in their default configurations raises an alert. This is not a flaw in how these tools are implemented; it is a direct consequence of the signature-based paradigm. ICMP tunnelling abuses a legitimate protocol in a non-standard way, producing no pattern that any signature database recognises. A behavioural system could flag the unusual size distribution and timing of those packets without any prior knowledge of the technique.

Additional limitations compound this fundamental problem. Rule sets require continuous maintenance as new threats emerge, which demands dedicated security staff. As TLS adoption grows, payload-level inspection becomes increasingly ineffective, leaving only metadata accessible. Broad rule sets also generate large numbers of false positive alerts, overwhelming analysts and causing genuine threats to be buried in noise.

3 AI-Based Anomaly Detection

3.1 Core Principle

The central insight behind anomaly-based detection is that normal network behaviour, while complex, is learnable. A well-configured enterprise network has predictable traffic patterns: consistent peak hours, a stable set of communicating hosts, characteristic protocol distributions, and typical flow sizes. An AI-based Anomaly Detection System (AI-ADS) trains a model on this normal traffic and then evaluates new flows against it. Flows that fall outside the learned normal region receive a high anomaly score and trigger an alert [1].

This approach involves a fundamental trade-off. A tight model of normal behaviour detects subtle deviations but also flags legitimate unusual events, such as a software deployment or a large backup, producing false positives. A looser model generates fewer alerts but may miss attacks that stay close to the boundary of normal behaviour. Choosing the right threshold is one of the central engineering challenges in deploying any AI-ADS.

3.2 Detection Pipeline

A typical AI-ADS processes traffic through the following stages.

Data collection. Raw packets are captured from a network tap or span port and parsed into structured records. Common representations include individual packets, reassembled TCP flows, or aggregated NetFlow records summarising per-connection statistics over a fixed time window.

Feature extraction. Numerical features are computed from each record. Typical flow-level features include flow duration, byte and packet counts in each direction, mean and variance of packet inter-arrival times, TCP flag combinations, and the number of distinct destination ports contacted per source address.

Preprocessing. Features are normalised so that high-magnitude values do not dominate distance calculations. Class imbalance between normal and attack samples is addressed using techniques such as SMOTE or undersampling.

Model training. The model is trained on a dataset of normal traffic, and in supervised settings also on labelled attack traffic. It learns a compact representation of the normal feature distribution.

Scoring and thresholding. For each new flow, the model produces an anomaly score. A configurable threshold determines whether the flow is classified as normal or anomalous, and is tuned on a validation set to achieve an acceptable balance between false positive and false negative rates.

Alerting and feedback. Anomalous flows are forwarded to the Security Operations Centre (SOC) for review. Confirmed attacks are added to the training set so the model can adapt over time.

3.3 Advantages Over Signature-Based Systems

The primary advantage is coverage of unknown threats. A zero-day exploit that disrupts normal traffic characteristics will be flagged even if its signature has never been documented. Encrypted attack traffic, which bypasses payload inspection entirely, can still trigger alerts through anomalous flow-level metadata. AI-ADS systems also reduce the burden of rule maintenance, since model parameters are learned from data rather than hand-crafted [3].

3.4 Challenges and Limitations

AI-based systems introduce their own problems. High false positive rates remain the most significant operational challenge: any legitimate deviation from the baseline, such as a network migration or a new service rollout, may trigger an alert. Normal behaviour also evolves over time as new applications are added and usage patterns change, a phenomenon called concept drift, which requires periodic retraining. Training data quality directly determines model quality; if the training period contains attack traffic, or if the dataset is not representative of the production network, the baseline itself is corrupted. Deep learning models also raise interpretability concerns: when a model flags a flow, it may be difficult to explain why in terms that an analyst can act on.

4 AI and Machine Learning Methodologies

A range of machine learning techniques has been applied to network anomaly detection. This section surveys the most important categories: classical machine learning algorithms, deep learning architectures, and clustering-based methods.

4.1 Classical Machine Learning

4.1.1 Isolation Forest

Isolation Forest [2] is an unsupervised ensemble method built on the observation that anomalies are few in number and structurally different from the majority of data. The algorithm builds a collection of binary decision trees by repeatedly partitioning the feature space using randomly chosen split features and split values. An anomalous point, lying far from the bulk of the data, ends up isolated in its own leaf after very few partitions. A normal point, embedded in a dense region, requires many more partitions to be separated. The average depth at which a point is isolated across all trees is its anomaly score: a shallow average depth corresponds to a high anomaly score, and a deep one corresponds to normal behaviour.

Isolation Forest is well-suited to detecting volumetric anomalies and isolated traffic spikes. Its linear time complexity makes it one of the most computationally efficient options for high-speed network monitoring, and it requires no labelled attack data, making it straightforward to deploy on a new network with no prior attack history.

4.1.2 One-Class Support Vector Machines

The One-Class SVM maps training data into a high-dimensional feature space via a kernel function and finds the smallest enclosing boundary around the training points. Any new sample that falls outside this boundary is classified as an anomaly. For network traffic, this approach is applied to flow-level feature vectors where normal flows cluster compactly. The RBF (radial basis function) kernel is most common because it can model non-linear boundaries. The main drawback is scalability: training cost grows quadratically with dataset size, making it impractical for large traffic volumes without subsampling.

4.1.3 Random Forest and Ensemble Methods

In supervised settings where labelled data for both normal and attack traffic is available, ensemble classifiers such as Random Forest and gradient-boosted trees (XGBoost, LightGBM) deliver strong performance [3]. Random Forest builds an ensemble of decision trees, each trained on a bootstrap sample with a random feature subset considered at each split, and aggregates predictions by majority vote. The feature importance scores it produces are particularly useful in security contexts, revealing which traffic features, such as byte count, flow duration, and TCP flag patterns, are most discriminative for separating normal traffic from attacks. The limitation of supervised methods is their dependence on labelled training data, which is expensive to produce and becomes stale as attacks evolve.

4.2 Deep Learning Approaches

4.2.1 Autoencoders

An autoencoder is a neural network trained to compress its input into a low-dimensional representation and then reconstruct the original from that representation. Training minimises the difference between the input and the reconstructed output, known as the reconstruction error. When the network is trained exclusively on normal traffic, it learns to reproduce normal patterns accurately. Attack traffic, whose structure was absent during training, is reconstructed poorly, producing a high reconstruction error. This error serves directly as the anomaly score: flows above a chosen error threshold are flagged as anomalous.

Kitsune [4] applies this principle in production using an ensemble of small autoencoders, each trained on a semantically coherent subset of network features, with the individual reconstruction errors combined into a single score. The system detected a wide range of attacks, including ARP poisoning, OS scanning, and video injection, in real network traffic with low false positive rates, and updates its model incrementally as traffic patterns change.

4.2.2 Recurrent Neural Networks and LSTM

Network traffic is a time series: the sequence of flows from a host over time encodes information about its behaviour that is invisible when flows are treated as independent samples. Long Short-Term Memory (LSTM) networks are designed for exactly this kind of sequential data. An LSTM processes flows one at a time and maintains an internal memory state that retains relevant context from earlier in the sequence while selectively discarding what is no longer needed. This selective memory is controlled by learned gating mechanisms that determine what information to keep, update, and output at each step.

For anomaly detection, an LSTM is trained to predict the next expected flow in a sequence of normal traffic. If the actual next flow deviates significantly from the prediction, the difference is flagged as an anomaly [5]. This makes LSTM particularly effective against slow-and-low exfiltration, advanced persistent threats that maintain persistent footholds with minimal per-flow traffic, and insider threats whose anomalous patterns only become visible over extended observation periods.

4.2.3 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) can be applied to network traffic by representing a window of packet bytes as a two-dimensional grid and treating it like an image. The convolutional filters learn local patterns in byte sequences characteristic of specific protocols or attack payloads. CNNs have shown strong results in malware traffic classification and in detecting attacks embedded in application-layer payloads. Hybrid CNN-LSTM architectures, which apply convolution to local feature windows and then feed the extracted features into a recurrent layer, have achieved top benchmark performance on datasets including NSL-KDD and CICIDS-2017 [9].

4.3 Clustering-Based Approaches

Clustering algorithms group data by similarity. Points that do not fit into any cluster, or that belong to very small sparse clusters, are natural anomaly candidates.

k-Means partitions data into a fixed number of clusters by iteratively assigning each point to its nearest centroid and recomputing centroids. After training on normal traffic, the anomaly score of a new flow is its distance to the nearest centroid. The method is fast and interpretable but requires the number of clusters to be set in advance and assumes roughly spherical cluster shapes.

DBSCAN identifies clusters as dense regions separated by lower-density areas and explicitly labels any points outside a dense region as noise, which are the anomaly candidates. It requires no predetermined number of clusters, handles arbitrary shapes, and directly represents the concept of an outlier. Its main limitation is sensitivity to the density threshold parameter and reduced performance in very high-dimensional feature spaces.

4.4 Comparison of Methodologies

Table 1 summarises the key properties of the methods above. No single approach dominates across all dimensions; the right choice depends on data availability, computational budget, interpretability requirements, and the expected threat profile.

Table 1: Comparison of AI/ML Methods for Network Anomaly Detection

Method	Supervision	Scalability	Temporal	Interpretability
Isolation Forest	Unsupervised	High	No	Medium
OC-SVM	Unsupervised	Low	No	Medium
Random Forest	Supervised	High	No	High
Autoencoder	Unsupervised	Medium	No	Low
LSTM	Unsupervised	Medium	Yes	Low
CNN	Supervised	Medium	Partial	Low
k-Means	Unsupervised	High	No	High
DBSCAN	Unsupervised	Medium	No	High

5 Applications, Discussion, and Mitigations

5.1 Benchmark Datasets and Evaluation

Evaluating anomaly detection systems requires representative datasets. Table 2 lists the benchmarks most frequently used in the literature.

Table 2: Commonly Used Benchmark Datasets

Dataset	Year / Origin	Key Characteristics
KDD Cup 99	1999, DARPA	First large benchmark; criticised for redundant records and unrealistic traffic [7]
NSL-KDD	2009, Univ. of New Brunswick	Cleaned version of KDD Cup 99; four attack categories
CICIDS-2017	2017, Canadian Inst. for Cybersecurity	Realistic modern traffic; DDoS, brute force, web attacks; flow-level labels [6]
UNSW-NB15	2015, UNSW Canberra	Nine attack families; widely used for deep learning evaluation [8]

Standard evaluation metrics are Precision ($TP/(TP + FP)$), Recall ($TP/(TP + FN)$), F1-Score, and False Positive Rate ($FP/(FP + TN)$). In production, the false positive rate often matters most: even a 1% false positive rate on a network carrying one million flows per hour produces ten thousand spurious alerts per hour, an unmanageable workload for any SOC team.

A widely cited concern raised by Sommer and Paxson [10] is that benchmark performance and operational performance frequently diverge. Models trained on laboratory datasets degrade noticeably when deployed on real networks because no public dataset captures the full diversity

and dynamism of live traffic. This is why training on data from the target network, rather than relying on pre-trained models from academic settings, is standard practice in serious deployments.

5.2 Enterprise Network Security

In a corporate environment, AI-ADS components are typically integrated into a SIEM platform that aggregates logs from firewalls, endpoint tools, authentication systems, and network sensors. The anomaly detection layer processes flow records in near real time, assigns risk scores, and forwards high-confidence alerts to the SOC.

A practical deployment uses a tiered approach: a fast, lightweight model such as Isolation Forest performs first-pass filtering of all traffic, and flows it flags are passed to a more expensive LSTM or autoencoder for deeper analysis. This keeps overall latency low while applying powerful models to the traffic that warrants closer attention [4].

User and Entity Behaviour Analytics (UEBA) extends this logic from network flows to individual accounts and hosts. By modelling the normal behaviour of each user, including login times, accessed resources, and volumes of data transferred, UEBA systems can detect compromised credentials and insider threats that generate no obvious network-level anomaly.

5.3 National Defence and Military Networks

Military and government networks face adversaries that are more resourced and more strategically motivated than in the commercial sector. Nation-state actors conduct sustained, targeted campaigns specifically designed to evade detection. Advanced Persistent Threats (APTs) operate at low traffic volumes over long time horizons, using legitimate protocols and valid credentials. Signature-based systems are ineffective against them by design.

LSTM-based detection is particularly relevant for APTs because these threats are defined by their temporal behaviour: slow beaconing to a command-and-control server, gradual lateral movement across network segments, or persistent low-rate exfiltration over weeks. No individual flow is anomalous; the pattern only becomes visible when traffic sequences are modelled over extended time windows [5].

Military networks also include air-gapped systems and classified enclaves with no direct internet connectivity. In these environments, detection focuses on internal patterns: unexpected connections between network segments, unusual access to sensitive data repositories, or the use of removable media. The very low tolerance for both false positives and false negatives in classified settings makes interpretable models and human-in-the-loop review essential.

5.4 Critical Infrastructure Protection

Critical infrastructure, covering power grids, water treatment facilities, transportation networks, and financial systems, increasingly depends on networked Operational Technology (OT) and Industrial Control Systems (ICS) such as SCADA platforms. These systems were historically air-gapped and ran proprietary protocols such as Modbus and DNP3, but growing connectivity requirements have eroded that isolation, and legacy equipment cannot be easily patched.

AI-based anomaly detection for ICS networks must account for the specific characteristics of OT traffic: low bandwidth, highly repetitive communication cycles, strict timing requirements, and a small, stable set of communicating devices. This regularity is actually an advantage for detection; the baseline is easy to establish and even minor deviations are suspicious.

Autoencoder-based methods trained on repetitive ICS protocol sequences have shown low false positive rates in this context.

The 2015 and 2016 attacks on Ukraine's power grid demonstrate what is at stake. The Sandworm threat actor used coordinated network intrusion to disrupt SCADA systems and cause widespread blackouts. An AI-ADS capable of detecting the anomalous reconnaissance and lateral movement that preceded those attacks would have been invisible to any signature-based tool, highlighting the practical value of behavioural detection in protecting critical infrastructure [3].

5.5 Advanced Persistent Threat Detection

APTs are the hardest problem in network anomaly detection because they are engineered to blend in. Common evasion methods include low-and-slow exfiltration, where data is moved out over weeks in small increments that keep per-flow volumes within the normal range; living-off-the-land techniques that abuse built-in system tools to generate traffic indistinguishable from legitimate administration; and domain fronting that tunnels command-and-control traffic through trusted cloud services.

Effective APT detection therefore requires combining network anomaly detection with host-based telemetry, DNS query analysis, TLS certificate inspection, and temporal correlation across multiple data sources. Graph-based anomaly detection, which models the network as a graph of communicating entities and identifies anomalous structural changes such as new edges or unexpected shifts in connectivity, is well-suited to detecting the lateral movement phase of an APT campaign and is an active area of current research.

5.6 Mitigation Strategies and Best Practices

Several operational practices are essential for deploying an AI-ADS reliably.

Hybrid detection. Signature-based and anomaly-based detection are complementary. Signature tools handle known threats with high precision and low latency; anomaly tools cover unknown ones. Neither is sufficient alone [10]. A layered architecture combining both is the standard recommendation.

Continuous retraining. Models must be updated regularly to account for concept drift. In dynamic networks, online learning methods that update model parameters incrementally with each new flow are preferable to periodic batch retraining, since they adapt in real time without requiring a scheduled maintenance window.

Threshold calibration. Thresholds must be tuned on data from the specific deployment network, not on a public benchmark. The acceptable false positive rate differs greatly between a small company and a military network, and the threshold must reflect the operational context.

Contextual alerting. Raw anomaly scores are not actionable. Alerts should be enriched with source host identity, affected service, and recent activity history to reduce analyst triage time and improve response accuracy.

Federated learning. In multi-organisation or classified settings where raw traffic cannot be centralised, federated learning allows models to be trained collaboratively without sharing data, satisfying both operational security requirements and data protection regulations such as GDPR.

Adversarial robustness. A sophisticated adversary may probe an AI-ADS to map its decision boundary and craft traffic that evades it. Training with adversarial examples and using

ensemble architectures that aggregate multiple models both improve resilience against deliberate evasion.

5.7 Open Challenges

Several problems continue to limit practical adoption. Public evaluation datasets are now several years old and do not reflect contemporary encrypted, high-speed traffic or recent attack techniques. Deep learning models remain largely opaque, and meeting demands for explainability without sacrificing detection performance is an unsolved problem. At 100 Gbps and above, even efficient models face throughput constraints that require hardware acceleration. Models trained on one organisation's traffic also generalise poorly to another's, making transfer learning an important open research direction.

6 Conclusion

Signature-based intrusion detection systems are a practical and widely deployed tool for identifying known threats. The demonstrated failure of Snort and Suricata to detect ICMP tunnelling in default configurations is representative of a broader limitation: any system whose coverage is bounded by a fixed rule catalogue will fail against novel attacks.

AI-based anomaly detection addresses this directly. By learning a model of normal traffic and flagging departures from it, these systems can identify zero-day exploits, covert channels, and slowly progressing intrusions that produce no matchable signature. Isolation Forest provides efficient unsupervised detection of volumetric anomalies. Autoencoders capture the statistical structure of normal traffic and flag flows with high reconstruction error without needing labelled attack data. LSTM networks model the temporal dimension of behaviour, making them the natural choice for detecting APTs and insider threats whose patterns only emerge over extended observation. Clustering methods offer interpretable, lightweight alternatives for environments where explainability is critical.

No single method is best in every context. The most effective deployments combine signature matching for known threats, anomaly scoring for unknown ones, and temporal modelling for persistent campaigns, all backed by continuous retraining and threshold calibration on local traffic. In high-stakes environments from enterprise SOCs to military networks and industrial control systems, this layered approach represents the current state of practice. As machine learning models continue to improve and hardware costs fall, anomaly-based detection will occupy an increasingly central role in network security architecture.

References

[1] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–58, 2009.

[2] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Mining (ICDM)*, 2008, pp. 413–422.

[3] A. L. Buczak and E. Guven, "A survey of data mining and machine learning methods for cyber security intrusion detection," *IEEE Communications Surveys & Tutorials*, vol. 18, no. 2, pp. 1153–1176, 2016.

[4] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, "Kitsune: An ensemble of autoencoders for online network intrusion detection," in *Proc. Network and Distributed System Security Symposium (NDSS)*, 2018.

[5] P. Malhotra, L. Vig, G. Shroff, and P. Agarwal, "Long short term memory networks for anomaly detection in time series," in *Proc. 23rd European Symposium on Artificial Neural Networks (ESANN)*, 2015, pp. 89–94.

[6] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. 4th Int. Conf. Information Systems Security and Privacy (ICISSP)*, 2018, pp. 108–116.

[7] M. Tavallaee, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the KDD CUP 99 data set," in *Proc. IEEE Symp. Computational Intelligence for Security and Defense Applications (CISDA)*, 2009, pp. 1–6.

[8] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems," in *Proc. Military Communications and Information Systems Conf. (MilCIS)*, 2015, pp. 1–6.

[9] R. Vinayakumar *et al.*, "Deep learning approach for intelligent intrusion detection system," *IEEE Access*, vol. 7, pp. 41525–41550, 2019.

[10] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *Proc. 31st IEEE Symp. Security and Privacy*, 2010, pp. 305–316.